A.3 Work, energy and power

Guiding questions

- How are concepts of work, energy and power used to predict changes within a system?
- How can a consideration of energetics be used as a method to solve problems in kinematics?
- How can transfer of energy be used to do work?

◆ **Work, *W*** The energy transfer that occurs when an object is moved with a force. More precisely, work done = force × displacement in the direction of the force.

◆ **Energy** Ability to do work.

In the last two topics, we have discussed movement (A.1), and how forces can change the motion of objects (A.2). In this topic (A.3) we will move on to introduce two very important, closely related, numerical concepts: **work** and **energy**. Together these provide the 'accounting system' for science, enabling explanations and useful predictions to be made.

Of course, 'work' and 'energy' are words in common use in everyday language, but in physics they have much more precise definitions.

Nature of science: Science as a shared endeavour

Same words, different meanings

The terms used in physics have very precise meanings – but the same words are often used differently in everyday life. There is a long list of such words, such as 'work', 'energy' and 'power', as well as the various meanings of 'conservation', 'law', 'momentum', 'pressure', 'stress', 'efficiency', 'heat', 'interference', 'temperature' and so on.

This ambiguity is a problem that all students must overcome when learning physics. For example, what is the connection, if any, between work = force × displacement and 'I have a lot of homework to do tonight'?

Work

SYLLABUS CONTENT

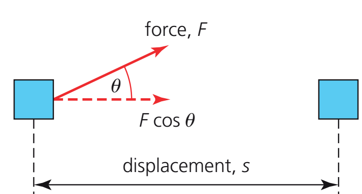
▶ Work, W, done on a body by a constant force depends on the component of the force along the line of displacement as given by: $W = Fs\cos\theta$.

Work done by constant forces

We say that work is done when any force moves an object: work is done *on* the object *by* the force. The work done, W, can be calculated by multiplying the displacement, s, by the component of the force acting in that direction, $F\cos\theta$.

See Figure A3.1 and note that θ is the angle between the force and the direction of motion.

work done, $W = Fs\cos\theta$ SI unit: Joule, J



■ **Figure A3.1** Work done by a force

One **joule** is defined to be the work done when a force of 1 N moves through a distance of 1 m.

◆ **Joule, J** Derived SI unit of work and energy. 1 J = 1 N m.

Commonly, a force acts in the same direction as the motion, in which case, the equation reduces to $W = Fs$.

WORKED EXAMPLE A3.1

Calculate how much work is done when a 1.5 kg mass is raised 80 cm vertically upwards.

Answer

The force needed to raise an object (at constant velocity) is equal to its weight (mg). The symbol h is widely used for vertical distances. (To avoid confusion, W will normally be used to represent work, and not weight.)

$W = Fs = mg \times h = 1.5 \times 9.8 \times 0.80 = 12\,\text{J}$

WORKED EXAMPLE A3.2

The 150 kg box in Figure A3.2 was pulled 2.27 m across horizontal ground by a force of 248 N, as shown.

a Determine how much work was done by the force.

b Suggest why it may make it easier to move the box if it is pulled in the direction shown by the dashed line.

c When the box was pulled at an angle of 20.0° to the horizontal, the force used to slide the box was 248 N. Calculate the work done by this force in moving the box horizontally the same distance.

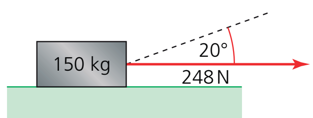


Figure A3.2 Box being pulled across the ground

Answer

a $W = Fs = 248 \times 2.27 = 563\,\text{J}$

b When the box is pulled in this direction, the force has a vertical component that helps reduce the normal contact force between the box and the ground. This will reduce the friction opposing horizontal movement.

c The force is not acting in the same direction as the movement. To calculate the work done we need to use the horizontal component of the 248 N force.

$W = F\cos 20° \times s = 248 \times 0.940 \times 2.27 = 529\,\text{J}$

It is important to realize that there are some surprising examples involving forces where *no* work is being done, as shown in Figures A3.3 and A3.4.

Figure A3.3 No work is being done on the weights at this moment

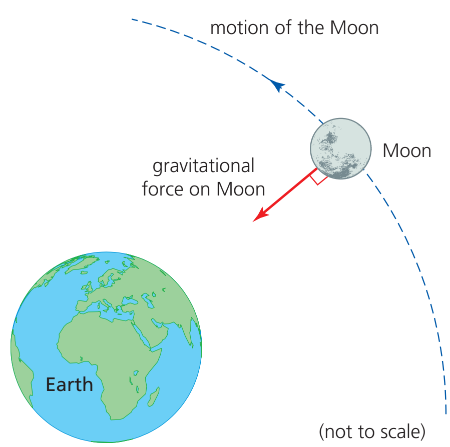


Figure A3.4 No work is done as the Moon orbits the Earth

In Figure A3.3 a large upwards force is being exerted on the stationary weights, but since there is no movement at that moment, no mechanical work is being done on the weights. However, work *is* being done in the weightlifter's muscles. In Figure A3.4, the Moon is moving *perpendicularly* to the force of gravity (cos 90° = 0), so there is no component of force in the direction of motion. A satellite in a similar circular path would not need to do any work, so that it would not need an engine to maintain the motion.

Work done by varying forces

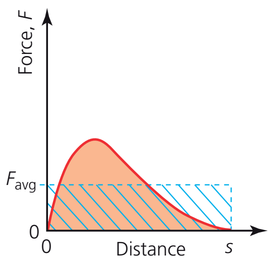


Figure A3.5 A force varying with distance

When making calculations with the equation $W = Fs\cos\theta$ we are assuming a single, constant value for the force, but in reality, forces are rarely constant. In order to calculate the work done by a *varying* force, we have to make an estimate of the *average* force involved. This may be best done with the help of a force–time graph, or a force–distance graph, as shown in Figure A3.5, which could represent, for example, the resultant force used to decelerate a car.

The horizontal dotted blue line represents the average force, as judged by eye.
Then, work done = $F_{avg} \times s$, which is the same as the rectangular area, and it is also equal to the area under the original curved line.

Work done is equal to the area under a force–distance graph.

WORKED EXAMPLE A3.3

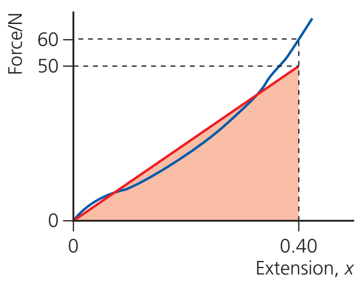


Figure A3.6 Force–extension (displacement) graph for rubber under tension

The blue line in Figure A3.6 represents the variations in force and extension as a rubber band was stretched. Estimate the work done as the force increased from zero to 60 N.

The work done is equal to the area under the graph. The red line has been drawn so that the area under it is the same as the area under the blue line (as judged by eye):

$$W = \frac{1}{2} \times 50 \times 0.40 = 10\,\text{J}$$

1 Calculate the work done to:
   a lift an 18 kg suitcase a vertical height of 1.05 m
   b push the same suitcase the same distance horizontally against an average frictional force of 37 N.

2 What was the magnitude of the average resistive force opposing the forward motion of a car if 2.3×10^6 J of work were done while maintaining a constant speed over a distance of 1.4 km?

3 In Figure A3.7 a gardener is pushing a lawnmower at a constant speed of 0.85 m s^{-1} with a force, F, of 70 N at an angle of 40° to the ground.
   a Calculate the component of F in the direction of movement.
   b What is the magnitude of the frictional force?
   c Determine the work done in moving the lawnmower for 3.0 s.

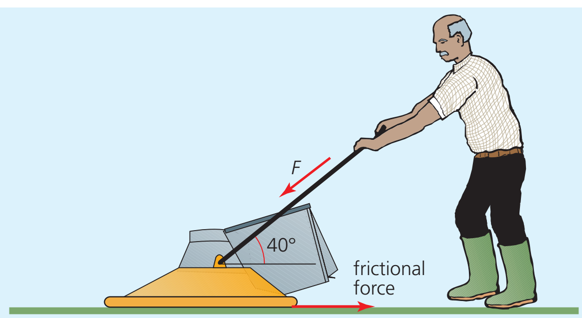


Figure A3.7 Gardener pushing a lawnmower

4 A spring which obeyed Hooke's law was stretched with a force which increased from zero to 24 N. The spring extended by 12 cm.
   a What was the average force used during the extension of the spring?
   b Calculate the work done on the spring.